$$D = \frac{\pi}{2} J(\Omega) \coth\left(\frac{\Omega}{2k_{\mathrm{B}}T}\right) = M\gamma_0 \Omega \frac{\Lambda^2}{\Lambda^2 + \Omega^2} \coth\left(\frac{\Omega}{2k_{\mathrm{B}}T}\right). \tag{5.48}$$